

On the Delay/Throughput-Security Tradeoff in Wiretap TDMA Networks With Buffered Nodes

Ahmed El Shafie[✉], *Senior Member, IEEE*, Naofal Al-Dhahir, *Fellow, IEEE*,
Zhiguo Ding[✉], *Senior Member, IEEE*, and
Ridha Hamila[✉], *Senior Member, IEEE*

Abstract—In this paper, we investigate the tradeoff between security and throughput and between security and queuing delay in wiretap time-division multiple access (TDMA) networks. We derive a simple relationship, characterized by a single key system parameter, between the stable-throughput region, where there are no perfect secrecy constraints on the data transmissions, and the secure stable-throughput region, where there are perfect secrecy constraints. We quantify the impact of the perfect secrecy constraints on the network's average queuing delay and propose a novel cross-layer security scheme for delay-limited applications. We establish an insightful link between computational security (i.e., upper-layer security implemented through cryptographic schemes) and physical-layer (information-theoretically proved) security. For the two-user case, we derive a closed-form expression for the network's minimum average queuing delay under the proposed security scheme and provide a relationship between the network's minimum queuing delay under perfect secrecy constraints and computational-only secrecy constraints. Moreover, we investigate the impact of cooperative jamming on achieving perfect secrecy, minimum network's queuing delay, and maximum throughput. We verify our theoretical findings through simulations.

Index Terms—Data buffers/queues, average queuing delay, uplink, secrecy stable-throughput region.

I. INTRODUCTION

CONFIDENTIALITY of legitimate users data from eavesdropping is very critical in wireless communications systems due to the broadcast nature of the medium. Internet-of-things (IoT) applications are growing significantly and,

at the same time, introducing new and significant security challenges [1], [2].

The security of classical encryption-based cryptographic schemes is based on the key assumption of limited computational power at the eavesdroppers [3]. To protect the transmitted data from eavesdropping security attacks, conventional computational security schemes use public-key cryptography to perform authentication and secret-key distribution and symmetric encryption. Integration of state-of-the-art algorithms such as the Rivest-Shamir-Adleman (RSA) and the Advanced Encryption Standard (AES) is considered secure for many applications because there are no known efficient attacks on public-key systems [4].

To achieve perfect communication secrecy in the presence of eavesdropping, physical-layer (PHY) security was proposed [5] to ensure information-theoretic confidentiality of data messages. In the PHY security paradigm, no computational restrictions are assumed on the eavesdroppers and mathematically provable information-theoretic statements can be made about the leaked information to the eavesdroppers. PHY security can be implemented as a part of a layered security solution where it provides an additional layer of security [3]. In PHY security, the two elements that control the secrecy rate, which is the amount of bits per channel use that can be decoded at the intended receiver without any information leakage at any eavesdropping receiver, are the legitimate channel rate and the eavesdropping channel rate. The difference between the two rates is the secrecy rate which can be increased by: 1) increasing the legitimate channel rate through increasing its signal-to-noise ratio (SNR) and/or 2) decreasing the eavesdropping channel rate through decreasing its SNR. The second approach can be realized through injecting artificial noise (AN) signals [6] to degrade the eavesdropping channels only which is also known as cooperative jamming [7].

Researchers have investigated the PHY security of several channel fading models such as quasi-static fading and ergodic fading for the wiretap channel. The authors of [8], [9] investigated the secrecy outage probability (SOP) for quasi-static fading channels and demonstrated that information-theoretic security is achievable in the presence of fading alone, due to the fluctuations of the channel gain, even when the eavesdropping receivers experience better average SNR than the intended receivers. On the other hand, the ergodic channel model was investigated in [10]–[12] where the authors

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A. El Shafie is with Qualcomm Inc., San Diego, CA 92121 USA (e-mail: aelshafie@qti.qualcomm.com).

N. Al-Dhahir is with Department of Electrical and Computer Engineering, The University of Texas at Dallas, Richardson, TX 75080 USA (e-mail: aldhahir@utdallas.edu).

Z. Ding is with the School of Electrical and Electronic Engineering, The University of Manchester, Manchester 46962, U.K. (e-mail: zhiguo.ding@manchester.ac.uk).

R. Hamila is with the Department of Electrical Engineering, Qatar University, Doha, Qatar (e-mail: hamila@qu.edu.qa).

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derived the channel secrecy rates for different channel state information (CSI) knowledge and derived transmit power allocation schemes. A detailed survey of PHY security schemes for various wireless networks is found in [13].

Investigating the PHY security of multiple-access systems has received increased attention recently [14]–[17]. In [14], the authors suggested a modification to the well-known slotted-ALOHA scheme where each legitimate data source node either transmits its data signal or decides to act as a cooperative jamming node according to a certain probability. The authors of [15] investigated random networks where the cooperative jammers and the eavesdroppers were assumed to be distributed according to independent two-dimensional homogeneous Poisson point processes (PPP). In [16], the authors investigated the secure-stable throughput region, which is defined as the throughput region under perfect secrecy and stable data queues, and delay-requirement region, which is defined as the region where all users' queuing-delay requirements are satisfied, assuming knowledge of the eavesdropper's instantaneous CSI and distributed jamming by the source nodes that are not selected for data transmission. In [17], the authors investigated PHY security of single-carrier frequency-division multiple-access (SC-FDMA) where time-domain AN is injected by the legitimate nodes using the degrees of freedom provided by the cyclic prefix.

Motivated by the importance of the computational and physical security paradigms and the fact that using one of them is not enough to achieve both perfect security and minimum queuing delay (or maximum throughput), we investigate in this paper the impact of integrating both approaches in a layered manner on the system's security-throughput-delay tradeoff. In the PHY security literature, if the eavesdropper's instantaneous CSI is known at the transmitting node, the transmitter knows the security status of the transmission (i.e., perfectly- or not perfectly-secured) and will only transmit during perfectly-secured time slots. Although this is the best approach in terms of security and user's confidentiality, it results in excessive queuing delays and high throughput loss. Alternatively, if the system employs both types of security approaches such that some of the not perfectly-secured time slots are secured using computationally-secured cryptographic-based schemes, the performance (in terms of delay or throughput) will be improved significantly.

In this paper, we consider a network with an arbitrary number of buffered legitimate users sharing the channel resources to communicate with their base station using a time-division multiple-access (TDMA) scheme in the presence of eavesdropping security attacks. The communication time is divided into slots. Since users access the channel using a TDMA scheme, only one legitimate user is allowed to transmit its data in a given time slot. We address the security challenges of such systems and analyze the throughput and queuing delay related performance metrics. The system's design parameters are optimized to achieve the users' quality-of-service (QoS) requirements. We emphasize here that, although the study in this paper focuses on the uplink scenario of a TDMA network as a case study, the presented ideas up to Section III are valid for other networks as it will be highlighted at the

end of Section III. TDMA is widely adopted multiple-access scheme in many existing networks such as the GSM cellular networks, Bluetooth personal area networks, IEEE 802.16a WiMax broadband wireless access networks, and 5G networks. Hence, by considering the general framework of TDMA networks, our work covers many existing networks.

The contributions of this paper are summarized as follows.

- We investigate the relationship between the reliable-only stable-throughput region, where security is not constrained to be perfect or only computational secrecy is implemented, and the secure stable-throughput region, where security is constrained to be perfect, for TDMA systems with node buffers. We derive a simple transformation between the two regions in terms of a single key system parameter and quantify the impact of jamming to reduce the gap between the two regions.
- We analyze the tradeoff between throughput and security and between queuing delay and security. We propose a new joint physical-layer and upper-layer secure scheme to increase the throughput and reduce the network's average queuing delay. The main idea is to relax the perfect secrecy constraint and to rely on computational security in a fraction of the communication time slots where the transmissions are known to be imperfectly secured.
- For the two-user case, we derive a closed-form expression for the network's minimum average queuing delay under the proposed security scheme and provide a relationship between the network's minimum queuing delay under perfect secrecy constraints and computational-only secrecy constraints.
- We quantify the impact of cooperative jamming on achieving both perfect secrecy and minimum queuing delay or maximum throughput. We prove that cooperative jamming reduces the gap between the reliable-only stable-throughput region and the secure stable-throughput region and that it achieves the lowest network's average queuing delay under perfect secrecy.
- For our considered TDMA system, we derive the optimal time slot assignment probabilities in closed-form for the cases of throughput and secure throughput maximization and the network's average queuing delay minimization.

Notation: $(\cdot)^*$ denotes complex-conjugate operation. $(\cdot)^T$ denotes vector transpose. $\|\cdot\|$ denotes the Euclidean norm of a vector. $|\cdot|$ denotes either absolute value or set cardinality depending on the context in which it is used. $\mathbb{E}\{\cdot\}$ denotes statistical expectation. $\mathbf{0}$ denotes the all-zero matrix/vector and its size is understood from the context. $[\cdot]^+ = \max(\cdot, 0)$ denotes the maximum between the enclosed values in brackets and **zero**. $\overline{X} = 1 - X$. A list of the key variables is given in Table I.

II. SYSTEM MODEL

We assume a generic TDMA network, shown in Fig. 1, with N source nodes sharing the same time and frequency resources. The legitimate users, Alices, wish to communicate confidentially with their destination node, Bob, in the presence of one or more eavesdropping node(s), Eve(s). Our analysis is not limited by the number of eavesdroppers or how they are

TABLE I
LIST OF KEY VARIABLES

Symbol	Description	Symbol	Description
N	Number of legitimate users (Alices)	T	Time slot (coherence time) duration
W	Communication channel bandwidth	$\mathcal{R}$	Target secrecy rate
$\eta_{k,B}$	Average receive SNR at Bob from the k -th Alice's transmission	η_{J,E_m}	Average receive SNR at the m -th Eve from the cooperative jammer's transmission
κ	Thermal noise spectral density	ω_k	Probability of assigning the k -th Alice to a time slot
Q_k	Data queue (buffer) at the k -th Alice	μ_k	Average service rate of the k -th Alice data queue
λ_k	Mean arrival rate at data queue Q_k	M	Number of transmit antennas at the cooperative jammer
$\mathbb{R}_{n_1,n_2}$	Instantaneous channel rate of the $n_1 - n_2$ link	R_k^{sec}	Instantaneous secrecy rate of the k -th Alice
h_{n_1,n_2}	Channel coefficient of the $n_1 - n_2$ link	β_{n_1,n_2}	Channel gain of the $n_1 - n_2$ link, i.e., $ h_{n_1,n_2} ^2$
f_k^{sec}	Probability of no secrecy outage of the $k - B$ link	f_k^{conn}	Probability of connection (no link outage) of the $k - B$ link
d_{n_1,n_2}	Distance between node n_1 and node n_2	μ_k^{sec}	Secure throughput of the k -th Alice

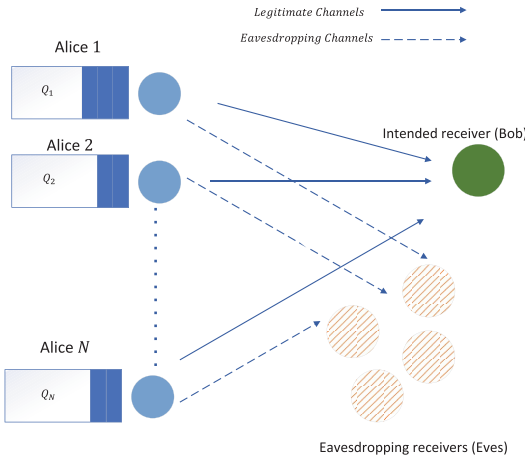


Fig. 1. The considered wireless network model where the k -th Alice has a queue/buffer Q_k with mean arrival rate λ_k . The number of users is N and they communicate with their intended receiver Bob in the presence of the eavesdropping nodes, Eves.

distributed in the network under consideration. Unless otherwise stated explicitly, Alices, Bob, and Eves are assumed to be equipped with single antennas. For simplicity, we assume that the Alices are numbered as $1, 2, \dots, N$. In our analysis, we use the subscripts E, B, and k to denote Eve, Bob, and the k -th Alice, respectively. We assume Rayleigh flat-fading channels. The channel coefficient (gain) between node n_1 and node n_2 , denoted by h_{n_1,n_2} ($\beta_{n_1,n_2} = |h_{n_1,n_2}|^2$), remains constant during a time slot, but it changes identically and independently from one time slot to another. The channel coefficient h_{n_1,n_2} is modeled as a circularly-symmetric Gaussian random variable with zero mean and unit variance. The distance between node n_1 and node n_2 is d_{n_1,n_2} and, hence, the pathloss is $d_{n_1,n_2}^{-\gamma}$, where γ is the pathloss exponent [8]. If a signal is transmitted from node n_1 with average power P , it will be received at

node n_2 with average power $d_{n_1,n_2}^{-\gamma} P$. The received signal at a receiver is perturbed by zero-mean additive white Gaussian noise (AWGN) with power spectral density κ . Time is slotted and a time slot duration is T seconds [18], [19]. Since a TDMA scheme is adopted, the k -th Alice is allocated a time slot for data transmission with probability $0 \leq \omega_k \leq 1$, where $\sum_{k=1}^N \omega_k \leq 1$ [18], [19]. A system designer can optimize the time-slot assignment probabilities ($\omega_1, \omega_2, \dots, \omega_N$) to meet different types of the users' QoS requirements (e.g., throughput, queue stability, queuing delay). The k -th Alice's average transmit power level is P_k . Throughout this paper, whenever perfect secrecy is required, we add the word "secure" to the throughput and the stable-throughput region.

A. Queue Assumptions

The k -th Alice is assumed to maintain a buffer Q_k to store her incoming traffic. The data arrival process at the k -th Alice is Bernoulli with mean $0 \leq \lambda_k \leq 1$ packets/slot [18]. Although the Bernoulli arrival model is simple, it captures the sporadic and random nature of data packet arrivals [20]. More importantly, in the analysis of the considered discrete-time queuing systems, Bernoulli arrivals see time averages (BASTA). This property is equivalent to the Poisson arrivals see time averages (PASTA) property in continuous-time queuing systems [21].¹ If $\lambda_k = 1$ packets/slot, the k -th Alice admits one data packet in each time slot.

B. Instantaneous Secrecy Rates

Assume that all the feedback and control signals within a time slot have a duration of T_o . Hence, the data transmission

¹For continuous time queues, Poisson arrivals see time averages (PASTA) property states that the probability of the state as seen by an outside random observer of the queuing system is the same as the probability of the state seen by an arriving customer. The same property holds in discrete-time queues with Bernoulli arrivals [20]–[22]. This property is important to analyze the average performance of a queuing system, especially the queuing delays.

time is $T - T_o$. Assuming that the packet size of a source node contains D bits and the communication channel bandwidth is W Hz, then the data transmission rate is $\mathcal{R} = \frac{D}{TW(1-\frac{T_o}{T})}$. The secrecy outage event occurs in a time slot when the transmission rate exceeds the instantaneous secrecy rate. Letting $\mathbb{R}_{n_1, n_2}$ denote the instantaneous channel rate of the $n_1 - n_2$ link, the secrecy rate of the k -th Alice is given by

$$R_k^{\text{sec}} = [\mathbb{R}_{k,B} - \mathbb{R}_{k,E}]^+ \leq \mathbb{R}_{k,B} \quad (1)$$

Since the data is transmitted in packets with target secrecy (data) rate $\mathcal{R}$, if $\mathcal{R}$ is greater than $\mathbb{R}_{k,B}$, then from (1) it is also greater than the secrecy rate R_k^{sec} . Hence, the data cannot be decoded reliably at Bob. For this reason, we assume that if $\mathcal{R}$ exceeds the instantaneous rate of the link connecting the Alice assigned for data transmission and Bob, Alice remains idle to conserve her battery energy.

C. Computational Security Coding and Wiretap Code Design

In conventional wireless systems, computationally-secure cryptographic schemes are used to secure the data transmissions. This paper does not assume a specific implementation of such schemes. We simply assume that an effective upper-layer secure scheme is employed and the system is computationally secured given this scheme. In addition to the cryptographic computational security coding used at the upper layers, we assume that Wyner's coding is implemented at the physical layer, which ensures the perfect (i.e. information-theoretic) secrecy. In a given communication time slot, one Alice transmits her data packet (encrypted using the cryptographic security scheme) to Bob. If Alice knows the instantaneous CSI of her link to Bob, she adaptively adjusts her transmission rate $\mathbb{R}_{k,B}$ to be as close as possible to the rate of the Alice-Bob link such that no outage events occur. Since we assume a quasi-static channel with fixed data transmission rate (where data is sent in packets), Alice uses a codebook $\mathcal{C}(2^{n\mathbb{R}_{k,B}}, 2^{n\mathcal{R}}, n)$ where $\mathcal{R}$ is the target secrecy rate (i.e. packet size in bits/sec/Hz), n is the codeword length, $2^{n\mathbb{R}_{k,B}}$ is the codebook size, and $2^{n\mathcal{R}}$ is the number of Alice's confidential data messages to transmit. The $2^{n\mathbb{R}_{k,B}}$ codewords are randomly grouped into $2^{n\mathcal{R}}$ bins. To transmit confidential message $m_d \in \{1, 2, \dots, 2^{n\mathcal{R}}\}$, Alice randomly selects a codeword from bin m_d .

In this paper, the data is at least computationally secured. In addition, the PHY security coding ensures that the system is perfectly secured over some period of time, or at all times, if the Alices know Eve's instantaneous CSI and transmit only when the channel is perfectly secured, i.e., when the instantaneous secrecy rate is higher than the target secrecy rate. If Eve's CSI is unknown at Alices, which is a more practical scenario, the data is perfectly secured when the target secrecy rate is lower than the instantaneous secrecy rate (no secrecy outage) or computationally secured otherwise.

III. THROUGHPUT ANALYSIS

First, we analyze the stable-throughput and secure stable-throughput regions of the considered network. Then, we analyze the queuing delays and the network's average queuing delay.

A. Queue Stability

From a network-layer point of view, a fundamental performance measure is the stability of the system's queues [18]. We aim at characterizing the stable-throughput region, which constitutes a theoretical limit on the data arrival rates while maintaining the queues stable, and the network's average queuing delay. The queue's stability condition is derived in Appendix A.

B. Reliable Stable-Throughput Region

Consider the case where the legitimate system either uses the computational security only schemes or is not concerned with security. In such cases, Alice transmits the packet at the head of her queue whenever the Alice-Bob link is connected (i.e., not in connection outage). A packet at the head of Q_k is delivered reliably if the k -th Alice is scheduled for data transmission, which occurs with probability ω_k , and the transmission is not in connection outage, which occurs with probability f_k^{conn} . Hence, the mean service rate of the k -th Alice's queue is given by

$$\mu_k = \omega_k f_k^{\text{conn}} \quad (2)$$

From Appendix A, the queue stability condition for the k -th Alice is $\lambda_k \leq \mu_k$. The stable-throughput region can be obtained from the queues' constraints that $\lambda_k \leq \mu_k$, for all k , and that $\sum_{k=1}^N \omega_k \leq 1$. That is, the region that satisfies both constraints is given by $\lambda_k \leq \mu_k = \omega_k f_k^{\text{conn}}$. Thus, $\omega_k \geq \frac{\lambda_k}{f_k^{\text{conn}}}$. Using the constraint that $\sum_{k=1}^N \omega_k \leq 1$, the stable-throughput region is given by

$$\mathcal{S}^{\text{conn}} = \left\{ (\lambda_1, \lambda_2, \dots, \lambda_N) : \sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{conn}}} < 1 \right\} \quad (3)$$

C. Secure Stable-Throughput Region Without Cooperative Jamming

When Eve's instantaneous CSI is known at Alice, Alice would only transmit when the transmissions are perfectly secured, i.e., when the instantaneous secrecy rate is higher than the target secrecy rate. Hence, $f_k^{\text{conn}} = f_k^{\text{sec}}$. That is, a packet departs the associated queue whenever its transmission is secured. Although this will result in perfect secrecy, the queuing delay will be higher than the cases when secrecy is ignored. Hence, the lowest network's queuing delay is achieved when there are no secrecy constraints.

The optimal values of ω_k are functions of f_k^{conn} and f_k^{sec} regardless of the actual security schemes implemented at the physical layer. That is, as long as the security scheme at the physical layer results in high values of f_k^{conn} and f_k^{sec} , the throughput will be high and the queuing delay will be low. Hence, one can abstract the physical layer into f_k^{conn} and f_k^{sec} . Specifically, when designing the optimal $\{\omega_k\}_{k=1}^N$, we do not care if the system is multiple-input multiple-output (MIMO), single-input single-output (SISO), or multiple-input single-output (MISO), or has single or multiple single-antenna (or multi-antenna) eavesdroppers. We are just concerned with the values of f_k^{conn} and f_k^{sec} .

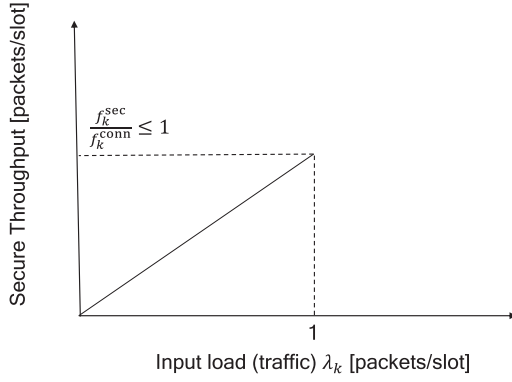


Fig. 2. The k -th Alice secure throughput versus input load, λ_k .

If a MIMO or MISO legitimate system is adopted with high transmit powers, it is obvious that the legitimate user can ensure that both f_k^{conn} and f_k^{sec} are close to 1. In this case, the optimal $\omega_k = \lambda_k$. The secure stable-throughput region is equal to the stable-throughput region and both are characterized by the set of arrivals $(\lambda_1, \lambda_2, \dots, \lambda_N)$ that satisfies $\sum_{k=1}^N \lambda_k < 1$. If the traffic loads are symmetric, which means that the arrival rate at each queue is $1/N$, then $\omega_k = \lambda_k = 1/N$ where N is the number of users in the network. The secure throughput will be equal to the queue service rate and both are equal to ω_k . Since $\sum_{k=1}^N \omega_k = 1$, the optimal value of ω_k is $1/N$ which makes sense since all users have the same requirements and traffic loads.

If there are no eavesdropping attacks, or if the cooperative jamming power is very high, then $f_k^{\text{sec}} = f_k^{\text{conn}}$. The secure throughput will be equal to the data throughput. The secure throughput, which is equal to the number of packets that are reliably decoded at Bob and completely secured from Eve, is given by

$$\mu_k^{\text{sec}} = \omega_k f_k^{\text{sec}} \Pr\{Q_k > 0\} \quad (4)$$

where $\Pr\{Q_k > 0\} = 1 - \Pr\{Q_k = 0\}$ and $\Pr\{Q_k = 0\}$ is given in Appendix A. Substituting with $\Pr\{Q_k > 0\} = \frac{\lambda_k}{\mu_k}$ and using Eqn. (4), we have

$$\mu_k^{\text{sec}} = \omega_k f_k^{\text{sec}} \frac{\lambda_k}{\mu_k} = \frac{f_k^{\text{sec}}}{f_k^{\text{conn}}} \lambda_k \quad (5)$$

which is independent of ω_k . Since the maximum arrival rate is $\lambda_k = 1$ packets/slot, the maximum secure throughput is $\frac{f_k^{\text{sec}}}{f_k^{\text{conn}}}$ packets/slot. In Fig. 2, we show the relation between the input traffic load and the secure throughput in (5) and emphasize that security constraints reduce the throughput.

Using the SOP expression in [8, Eqn. (7)], and substituting with our system's parameters, the no secrecy outage probability (NSOP) is given by

$$f_k^{\text{sec}} = \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} 2^{\mathcal{R}}} \exp\left(-\frac{2^{\mathcal{R}} - 1}{\eta_{k,B}}\right) \quad (6)$$

where $\eta_{k,B} = d_{k,B}^{-\gamma} \frac{P_k}{\kappa W}$ is the average receive SNR at Bob from the k -th Alice transmission. Eqn. (6) can be rewritten as

$$f_k^{\text{sec}} = \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} 2^{\mathcal{R}}} f_k^{\text{conn}} \stackrel{\text{def}}{=} \zeta_k f_k^{\text{conn}} \quad (7)$$

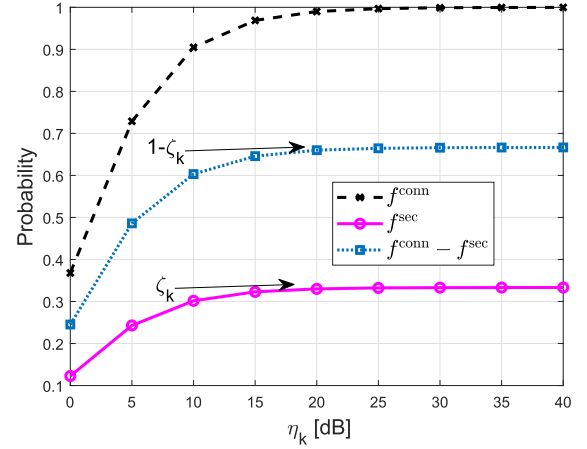


Fig. 3. The connection probability, the NSOP, and their difference versus $\eta_{k,B} = \eta_{k,E} = \eta_k$. The figure is generated with $\mathcal{R} = 1$ bits/sec/Hz.

where $f_k^{\text{conn}} = \exp\left(-\frac{2^{\mathcal{R}} - 1}{\eta_{k,B}}\right)$ is the probability that the direct link (i.e., the Alice-Bob link) is not in connection outage. The parameter ζ_k represents the reduction in the NSOP due to the presence of Eve or, equivalently, it is the increase of outage events due to the presence of Eve. When $d_{k,B} = d_B$ and $d_{k,E} = d_E$ for all k , we have

$$\zeta_k = \zeta = \frac{d_B^{-\gamma}}{d_B^{-\gamma} + d_E^{-\gamma} 2^{\mathcal{R}}} \quad (8)$$

where ζ is independent of the transmitting node. When $d_{k,B} = d_{k,E}$ for all k , then $\frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} 2^{\mathcal{R}}} = \zeta_0 = \frac{1}{1 + 2^{\mathcal{R}}}$.

As Alice's transmit power increases, $f_k^{\text{conn}} = \exp\left(-\frac{2^{\mathcal{R}} - 1}{\eta_{k,B}}\right)$ becomes closer to 1. However, the term $\zeta_k = \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} 2^{\mathcal{R}}}$ is independent of Alice's transmit power. Hence, eventually, the NSOP becomes constant at $f_k^{\text{sec}} = \zeta_k$. We demonstrate all these facts in Fig. 3 where we see that the connection probability is typically much larger than the NSOP. The figure is generated with $\mathcal{R} = 1$ bits/sec/Hz. One can interpret the connection probability and the NSOP as follows. Assume that we have a large number of time slots. A fraction of f_k^{conn} of those time slots can be used immediately if the system relies only on computational security or if there are no security constraints. However, only a fraction of f_k^{sec} can be used under the perfect security constraints. At $\eta_{k,B} = 25$ dB, under computational-only secrecy, the k -th Alice can transmit in all time slots and achieve almost outage-free transmissions. However, under the perfect secrecy constraints, only 33% of the time slots can be used. Moreover, the smallest value of ζ is $1/2$ and occurs when $\mathcal{R} = 0$. While the value of $\mathcal{R} = 0$ is not practical, it shows that the perfect secrecy constraint can make the throughput and the probability of secure packet delivery at Bob half its value when there are no eavesdropping security attacks or perfect secrecy constraints. One way to increase ζ is by introducing cooperative jamming to the eavesdropping channels, as will be explained shortly.

Substituting with (7) into (5), the secure throughput can be rewritten as

$$\mu_k^{\text{sec}} = \zeta_k \lambda_k \quad (9)$$

As shown above, ζ_k is equal for all users as long as the links have the same statistics, i.e., $\zeta_k = \zeta$. Hence,

$$\mu_k^{\text{sec}} = \zeta \lambda_k \quad (10)$$

and, by summing over the legitimate user index k , the secure sum throughput is given by

$$\sum_{k=1}^N \mu_k^{\text{sec}} = \zeta \sum_{k=1}^N \lambda_k \quad (11)$$

In addition, by dividing (7) over λ_k , taking the reciprocal, and summing over the legitimate user index k , we get

$$\sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{sec}}} = \frac{1}{\zeta} \sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{conn}}} \quad (12)$$

where $\sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{sec}}}$ is the secure-stable throughput region and $\sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{conn}}}$ is stable-throughput region. From (12), since the right hand side (RHS) $\sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{conn}}}$ represents the stable-throughput region of the system, we can conclude that the secure stable-throughput region, given by $\sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{sec}}}$, can be deduced directly from the stable-throughput region (with no eavesdropping) by dividing over ζ .

Using the relationship derived in (12), the secure stable-throughput region is given by

$$\mathcal{S}^{\text{sec}} = \left\{ (\lambda_1, \lambda_2, \dots, \lambda_N) : \sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{sec}}} < 1 \Rightarrow \sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{conn}}} < \zeta \right\} \quad (13)$$

where $0 \leq \zeta \leq 1$ which means that the sum $\sum_{k=1}^N \frac{\lambda_k}{f_k^{\text{conn}}}$ is decreased.

D. Secure Stable-Throughput Region With Cooperative Jamming

Now, we analyze the NSOP when there is a cooperative jammer. Assume that the cooperative jammer is equipped with M antennas. Denote the i -th ($i \in \{1, 2, \dots, M\}$) antenna at the cooperative jammer as J_i . Let $\mathbf{h}_{J-E} = [h_{J_1,E}, h_{J_2,E}, \dots, h_{J_M,E}]^T \in \mathbb{C}^{M \times 1}$ denote the channel vector between the cooperative jammer and Eve and $\mathbf{h}_{J-B} = [h_{J_1,B}, h_{J_2,B}, \dots, h_{J_M,B}]^T \in \mathbb{C}^{M \times 1}$ denote the channel vector between the cooperative jammer and Bob. The design of the **optimal** precoder at the cooperative jammer depends on the knowledge of Eve's instantaneous CSI. If Eve's instantaneous CSI is unknown at the cooperative jammer, the jammer designs the precoder based on its channels to Bob. In all cases, we maintain the condition that the interference due to AN is canceled at Bob.

Since Bob is the intended receiver, Alice can estimate the CSI of her link to Bob through information feedback from Bob. However, Eve's instantaneous CSI is either known or unknown. If it is known, as shown in [16], [23], Alice designs the precoder as a single vector that maximizes the instantaneous secrecy rate which is given by

$$\mathbf{p}^* = \frac{\mathbf{B}_p \mathbf{h}_{J-E}}{\|\mathbf{B}_p \mathbf{h}_{J-E}\|} \quad (14)$$

where $\mathbf{B}_p = \mathbf{I}_M - \frac{\mathbf{h}_{J-B} \mathbf{h}_{J-B}^*}{\|\mathbf{h}_{J-B}\|^2}$ and $\mathbf{I}_M$ denotes the $M \times M$ identity matrix. If Eve's instantaneous CSI is unknown, which is a more practical scenario, the cooperative jammer will design the AN to be orthogonal to the channel vector $\mathbf{h}_{J-B}$. That is, the cooperative jammer designs its AN-precoding matrix filter, denoted by $\mathbf{P}$, such that $\mathbf{h}_{J-B}^T \mathbf{P} = \mathbf{0}$. Since $\mathbf{h}_{J-B}^T$ is a $1 \times M$ vector, there are $M-1$ orthogonal directions to it. Hence, the cooperative jammer distributes its jamming power equally among those directions to hurt Eve since it does not know Eve's exact channel vector direction. If Eve's instantaneous CSI is known, then $\mathbf{P} = \mathbf{p}^*$.

Under cooperative jamming, the instantaneous secrecy rate of the k -th Alice transmission is given by

$$R_k^{\text{sec}} = \left[\log_2 \left(\frac{1 + \eta_{k,B} \beta_{k,B}}{1 + \frac{\eta_{k,B} \beta_{k,E}}{1 + \eta_{J,E} \|\mathbf{h}_{J-E}\|^2 \mathbf{P}^2}} \right) \right]^+ \quad (15)$$

where $\eta_{J,E} = d_{J,E}^{-\gamma} \frac{P_J}{N_s \kappa W}$ is the average receive SNR at Eve from the cooperative jammer's transmission with N_s denoting the number of AN symbols (which is equal to $M-1$ when Eve's instantaneous CSI is unknown and is equal to 1 when it is known) and $\mathbf{P}$ is the AN-precoding matrix filter with P_J denoting the cooperative jammer's transmit power. The NSOP under cooperative jamming is thus given by

$$\begin{aligned} f_k^{\text{sec}} &= \Pr\{\mathcal{R} \leq R_k^{\text{sec}}\} \\ &= \Pr\left\{ \mathcal{R} \leq \left[\log_2 \left(\frac{1 + \eta_{k,B} \beta_{k,B}}{1 + \frac{\eta_{k,B} \beta_{k,E}}{1 + \eta_{J,E} \|\mathbf{h}_{J-E}\|^2 \mathbf{P}^2}} \right) \right]^+ \right\} \end{aligned} \quad (16)$$

An intuitive approach to analyze the impact of jamming is to rewrite the expression in (6) in the presence of jamming. That is, for a given jamming SNR of $\eta_{J,E} \|\mathbf{h}_{J-E}^T \mathbf{P}\|^2$ at Eve, the NSOP is given by

$$f_k^{\text{sec}} \Big|_{\|\mathbf{h}_{J-E}^T \mathbf{P}\|^2} = \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} \frac{2\mathcal{R}}{1 + \eta_{J,E} \|\mathbf{h}_{J-E}^T \mathbf{P}\|^2}} f_k^{\text{conn}} \quad (17)$$

where $\cdot|_x$ means the probability given x . When $d_{k,B} = d_B$ and $d_{k,E} = d_E$ for all k , we get

$$f_k^{\text{sec}} \Big|_{\|\mathbf{h}_{J-E}^T \mathbf{P}\|^2} = \frac{d_B^{-\gamma}}{d_B^{-\gamma} + d_E^{-\gamma} \frac{2\mathcal{R}}{1 + \eta_{J,E} \|\mathbf{h}_{J-E}^T \mathbf{P}\|^2}} f_k^{\text{conn}} \quad (18)$$

Hence, we can control the SOP using $\eta_{J,E}$ and the AN-precoding matrix filter $\mathbf{P}$. In this case, again, the increase of outage events due to the presence of Eve under cooperative jamming, denoted by ζ_k^J , will be equal for all k , i.e., $\zeta_k^J = \zeta^J \forall k$, and is given by the expectation of $\frac{d_B^{-\gamma}}{d_B^{-\gamma} + d_E^{-\gamma} \frac{2\mathcal{R}}{1 + \eta_{J,E} \|\mathbf{h}_{J-E}^T \mathbf{P}\|^2}}$. The ratio ζ^J under cooperative jamming is monotonically non-decreasing with $\eta_{J,E}$ and by setting $\eta_{J,E}$ to infinity (high value), we can see that the gap between f_k^{conn} and f_k^{sec} becomes equal to 0 (i.e., $\zeta_k^J \rightarrow 1$). In general, under cooperative jamming, $\zeta_k \leq \zeta_k^J \leq 1$ with ζ_k denoting the increased outage due to eavesdropping in case of no cooperative jamming.

Assuming equal distances between the Alices and both Bob and Eve, i.e., $d_{k,B} = d_{k,E}$,

$$f_k^{\text{sec}} = \mathbb{E} \left\{ \frac{1}{1 + \frac{2^{\mathcal{R}}}{1 + \eta_{J,E} \|\mathbf{h}_{J-E}^{\top} \mathbf{P}\|^2}} \right\} f_k^{\text{conn}} \stackrel{\text{def}}{=} \zeta_k^J f_k^{\text{conn}} \quad (19)$$

From the above discussion, we can again write the relationship between the NSOP and the connection probability under cooperative jamming in the same form as Eqn. (7), namely

$$f_k^{\text{sec}} = \zeta_k^J f_k^{\text{conn}} \quad (20)$$

with $\zeta_k^J = \mathbb{E} \left\{ \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} \frac{2^{\mathcal{R}}}{1 + \eta_{J,E} \|\mathbf{h}_{J-E}^{\top} \mathbf{P}\|^2}} \right\}$ where the expectation is over the random variable $\|\mathbf{h}_{J-E}^{\top} \mathbf{P}\|^2$. Since $\mathbf{h}_{J-E}^{\top}$ is a zero-mean Gaussian random vector with unit-variance elements, the random variable $\|\mathbf{h}_{J-E}^{\top} \mathbf{P}\|^2$ is Chi-square with $2(M-1)$ degrees of freedom. At high number of cooperative jammer's transmit antennas, using the law of large numbers, $\|\mathbf{h}_{J-E}^{\top} \mathbf{P}\|^2 \rightarrow M$. Hence, $\zeta_k^J \rightarrow \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} \frac{2^{\mathcal{R}}}{1 + \eta_{J,E} M}}$. If Eve's instantaneous CSI is known at the cooperative jammer, $\eta_{J,E} = d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}$ and, hence, $\zeta_k^J \rightarrow \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} \frac{2^{\mathcal{R}}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} M}} \approx 1$

since M is large and $d_{k,E}^{-\gamma} \frac{2^{\mathcal{R}}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} M} \rightarrow 0$. This implies that the connection probability and the NSOP are equal and there is no impact of the eavesdropping attacks on the system's performance. If Eve's instantaneous CSI is unknown at the cooperative jammer, $\eta_{J,E} = d_{J,E}^{-\gamma} \frac{P_J}{(M-1)\kappa W}$ and, hence, $\zeta_k^J \rightarrow \frac{d_{k,B}^{-\gamma}}{d_{k,B}^{-\gamma} + d_{k,E}^{-\gamma} \frac{2^{\mathcal{R}}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}}}$. As the cooperative jamming transmit power increases, the term $d_{k,E}^{-\gamma} \frac{2^{\mathcal{R}}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}}$ becomes closer to zero. Hence, ζ_k^J becomes closer to 1.

For the case of multiple non-colluding eavesdroppers, the eavesdropper with the highest rate determines the instantaneous secrecy rate. Assume L eavesdropping nodes and let $h_{A_k-E_m}$ denote the channel coefficient between the k -th Alice and the m -th ($m \in \{1, 2, \dots, L\}$) Eve. In addition, we assume that $\mathbf{h}_{J-E_m}$ denotes the channel vector between the cooperative jammer and the m -th Eve and using the expression in (16), then the NSOP for given eavesdropping channels' realizations is given by

$$\begin{aligned} f_k^{\text{sec}} \Big|_{\mathcal{X}_{k,\tilde{m}}} &= \Pr \left\{ \mathcal{R} \leq \left[\log_2 \left(\frac{1 + \eta_{k,B} \beta_{k,B}}{1 + \eta_{k,B} \mathcal{X}_{k,\tilde{m}}} \right) \right]^+ \right\} \\ &= \Pr \left\{ \frac{2^{\mathcal{R}} (1 + \eta_{k,B} \mathcal{X}_{k,\tilde{m}}) - 1}{\eta_{k,B}} \leq \beta_{k,B} \right\} \\ &= \Pr \left\{ \frac{2^{\mathcal{R}} - 1 + (2^{\mathcal{R}} \eta_{k,B} \mathcal{X}_{k,\tilde{m}})}{\eta_{k,B}} \leq \beta_{k,B} \right\} \end{aligned} \quad (21)$$

where $\eta_{J,E_m} = d_{J,E_m}^{-\gamma} \frac{P_J}{(M-1)\kappa W}$, $\mathcal{X}_{k,\tilde{m}} = \max_{1 \leq m \leq L} \frac{\beta_{A_k,E_m}}{1 + \eta_{J,E_m} \|\mathbf{h}_{J-E_m}^{\top} \mathbf{P}\|^2}$ with $\tilde{m} = \underset{1 \leq m \leq L}{\operatorname{argmax}} \frac{\beta_{A_k,E_m}}{1 + \eta_{J,E_m} \|\mathbf{h}_{J-E_m}^{\top} \mathbf{P}\|^2}$. Since $\beta_{k,B}$ is an exponentially-distributed random variable with unit mean,

we get

$$\begin{aligned} f_k^{\text{sec}} \Big|_{\mathcal{X}_{k,\tilde{m}}} &= \exp \left(-\frac{2^{\mathcal{R}} - 1 + (2^{\mathcal{R}} \eta_{k,B} \mathcal{X}_{k,\tilde{m}})}{\eta_{k,B}} \right) \\ &= \exp \left(\frac{2^{\mathcal{R}} \eta_{k,B} \mathcal{X}_{k,\tilde{m}}}{\eta_{k,B}} \right) \exp \left(-\frac{2^{\mathcal{R}} - 1}{\eta_{k,B}} \right) \\ &= \exp(-2^{\mathcal{R}} \mathcal{X}_{k,\tilde{m}}) f_k^{\text{conn}} \end{aligned} \quad (22)$$

where $f_k^{\text{conn}} = \exp \left(-\frac{2^{\mathcal{R}} - 1}{\eta_{k,B}} \right)$ and L is the number of eavesdroppers. Hence,

$$f_k^{\text{sec}} = \mathbb{E} \left\{ \exp \left(-2^{\mathcal{R}} \max_{1 \leq m \leq L} \frac{\beta_{A_k,E_m}}{1 + \eta_{J,E_m} \|\mathbf{h}_{J-E_m}^{\top} \mathbf{P}\|^2} \right) \right\} f_k^{\text{conn}} \quad (23)$$

where the expectation in (23) is over the random variables β_{A_k,E_m} and $\|\mathbf{h}_{J-E_m}^{\top} \mathbf{P}\|^2$. Letting $\zeta_k^J = \mathbb{E} \left\{ \exp \left(-2^{\mathcal{R}} \max_{1 \leq m \leq L} \frac{\beta_{A_k,E_m}}{1 + \eta_{J,E_m} \|\mathbf{h}_{J-E_m}^{\top} \mathbf{P}\|^2} \right) \right\}$, the relationship between f_k^{sec} and f_k^{conn} follows Eqns. (7) and (20). Hence, the same conclusions about the relationship between the stable-throughput and the secure stable-throughput regions hold.

To obtain some insights of the cooperative jamming under multiple eavesdroppers scenario and its parameter $\zeta_k^J = \mathbb{E} \left\{ \exp \left(-2^{\mathcal{R}} \max_{1 \leq m \leq L} \frac{\beta_{A_k,E_m}}{1 + \eta_{J,E_m} \|\mathbf{h}_{J-E_m}^{\top} \mathbf{P}\|^2} \right) \right\}$, we consider the case of large M . When M is large $\|\mathbf{h}_{J-E_m}^{\top} \mathbf{P}\|^2 \rightarrow M$,

$$\begin{aligned} \zeta_k^J &= \mathbb{E} \left\{ \exp \left(-2^{\mathcal{R}} \max_{1 \leq m \leq L} \frac{\beta_{A_k,E_m}}{1 + \eta_{J,E_m} M} \right) \right\} \\ &= \mathbb{E} \left\{ \exp \left(-2^{\mathcal{R}} \max_{1 \leq m \leq L} \frac{\beta_{A_k,E_m}}{1 + d_{J,E_m}^{-\gamma} \frac{P_J}{\kappa W}} \right) \right\} \end{aligned} \quad (24)$$

with $\eta_{J,E_m} = d_{J,E_m}^{-\gamma} \frac{P_J}{(M-1)\kappa W}$. Assuming that all Eves are close to each other and have the same distances to Alice, or as a worst-case scenario for Alice, let all Eves' distances be equal to the Eve's distance with the minimum value among the Eves' distances, we get

$$\begin{aligned} \zeta_k^J &= \mathbb{E} \left\{ \exp \left(-2^{\mathcal{R}} \max_{1 \leq m \leq L} \frac{\beta_{A_k,E_m}}{1 + \eta_{J,E_m} M} \right) \right\} \\ &= \mathbb{E} \left\{ \exp \left(-2^{\mathcal{R}} \frac{\max_{1 \leq m \leq L} \beta_{A_k,E_m}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}} \right) \right\} \end{aligned} \quad (25)$$

where $d_{J,E}^{-\gamma} = \min_{1 \leq m \leq L} d_{J,E_m}^{-\gamma}$. As the cooperative jamming power increases significantly, regardless of the instantaneous value of $\max_{1 \leq m \leq L} \beta_{A_k,E_m}$, $\frac{\max_{1 \leq m \leq L} \beta_{A_k,E_m}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}} \rightarrow 0$. Hence, $\zeta_k^J \rightarrow 1$. To show that the quantity $\max_{1 \leq m \leq L} \beta_{A_k,E_m}$ does not impact the results, using the cumulative distribution function (CDF) of $\max_{1 \leq m \leq L} \beta_{A_k,E_m}$, we get

$$\Pr \left\{ \max_{1 \leq m \leq L} \beta_{A_k,E_m} < x \right\} = \left(1 - \exp(-x) \right)^L \quad (26)$$

since β_{A_k,E_m} is a unit-mean exponential-distributed random variable. Assuming that $x = 10$ and $L = 10$, we have

$\Pr \{ \max_{1 \leq m \leq 10} \beta_{A_k, E_m} > 10 \} = 1 - \Pr \{ \max_{1 \leq m \leq 10} \beta_{A_k, E_m} < 10 \} = 4.54 \times 10^{-4}$, which is very negligible when $d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}$ is higher than 15 dB.

Next, we find the minimum cooperative jamming transmit power to make ζ_k^J as close to 1 as possible, which implies that the eavesdropper has no impact on the system achievable performance. From (25), to make $\zeta_k^J \approx 1$, the condition $2^{\mathcal{R}} \frac{\max_{1 \leq m \leq L} \beta_{A_k, E_m}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}} \leq \delta$ needs to be satisfied, where δ is a very small number in the range of 10^{-3} . Hence, using (26) and substituting with $x = \delta 2^{-\mathcal{R}} \left(1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} \right)$, the probability $\Pr \left\{ 2^{\mathcal{R}} \frac{\max_{1 \leq m \leq L} \beta_{A_k, E_m}}{1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W}} \leq \delta \right\}$ is given by

$$\begin{aligned} & \Pr \left\{ \max_{1 \leq m \leq L} \beta_{A_k, E_m} < \delta 2^{-\mathcal{R}} \left(1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} \right) \right\} \\ &= \left(1 - \exp \left(-\delta 2^{-\mathcal{R}} \left(1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} \right) \right) \right)^L \\ &\approx \left(1 - L \exp \left(-\delta 2^{-\mathcal{R}} \left(1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} \right) \right) \right) \end{aligned} \quad (27)$$

where the last approximation holds from the fact that the exponent is very large. That is, we make use of the fact that $(1 - y)^L \approx 1 - Ly$ when y is much less than 1. To make the probability in (27) close to 1, $L \exp \left(-\delta 2^{-\mathcal{R}} \left(1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} \right) \right)$ must be very small, i.e., almost be equal to δ . From (27), the constraint on the cooperative jamming transmit power to achieve $\zeta_k^J \approx 1$ is

$$\delta 2^{-\mathcal{R}} \left(1 + d_{J,E}^{-\gamma} \frac{P_J}{\kappa W} \right) \geq -\ln \left(\frac{\delta}{L} \right) \Rightarrow P_J \geq \frac{\kappa W}{d_{J,E}^{-\gamma}} \left(\frac{\ln \left(\frac{L}{\delta} \right)}{2^{-\mathcal{R}} \delta} - 1 \right) \quad (28)$$

This interesting expression connects the cooperative jammer's transmit power with both the target secrecy rate and the number of eavesdroppers. If the cooperative jammer transmits its jamming signals with transmit power $\frac{\kappa W}{d_{J,E}^{-\gamma}} \left(\frac{\ln \left(\frac{L}{\delta} \right)}{2^{-\mathcal{R}} \delta} - 1 \right)$, $\zeta_k^J = 1$ and, hence, the NSOP will be equal to the connection probability.

E. Other Multiple-Access Systems

We emphasize here that, the analysis presented in this section is generally applicable as long as the queue's mean service rate can be written as in (2) and the secure throughput is written as in (4). For example, assuming a random multiple-access scheme where the k -th Alice accesses the channel randomly with probability q_k and that the users are backlogged with data packets, the probability that the k -th Alice transmission is decoded at Bob is given by

$$\mu_k = q_k \left(\prod_{\substack{r=1 \\ r \neq k}} (1 - q_r) \right) f_k^{\text{conn}} \quad (29)$$

where $q_k \left(\prod_{\substack{r=1 \\ r \neq k}} (1 - q_r) \right)$ is the probability that the user accesses the channel with no collision with other users.

The probability that a queue is nonempty follows the derived expression in Appendix A using the relevant μ_k in (29). The secure throughput of the k -th Alice is given by

$$\begin{aligned} \mu_k^{\text{sec}} &= q_k \left(\prod_{\substack{r=1 \\ r \neq k}} (1 - q_r) \right) f_k^{\text{sec}} \frac{\lambda_k}{q_k \left(\prod_{\substack{r=1 \\ r \neq k}} (1 - q_r) \right) f_k^{\text{conn}}} \\ &= \frac{f_k^{\text{sec}}}{f_k^{\text{conn}}} \lambda_k \end{aligned} \quad (30)$$

which is the same relationship derived for the TDMA system. Similar arguments can be made for other orthogonal multiple-access schemes such as code-division multiple-access (CDMA) and frequency-division multiple-access (FDMA) schemes. In addition, systems where a transmitting user is the node with the highest channel gain among the channels between the legitimate users and the base station, the node with the highest channel rate between the legitimate users and the base station, or, when Eve's instantaneous CSI is known at the transmitting nodes, the node with the highest instantaneous secrecy rate between the legitimate users and the base station. All those schemes will eventually result in a certain time-slot assignment probability to one of the users. When the communication links are statistically identical, under those schemes, a legitimate user is selected for transmission $1/N$ with N denoting the number of legitimate users.

F. Proposed Mixed Computational-Perfect Security Scheme

In this subsection, we propose a joint PHY and cryptographic security scheme. The system under consideration is assumed to be perfectly secured when the PHY security schemes are used, while the system is assumed to be partially secured when the PHY security schemes are not used and only the upper-layer cryptographic-based computational security schemes are used. This approach is motivated by the fact that relying only on perfect PHY secrecy would result in significant queuing delays, especially in scenarios where the eavesdroppers are at the same distance or closer to Alice than the intended receiver. If Eve's instantaneous CSI is perfectly known at transmitter, the perfectly secured time periods will be determined accurately and Alice can only transmit during those periods. However, this might lead to significant delays when the secure periods are short, as in the case described above.

Using time sharing between the NSOP and the connection probability, Alice can decide whether to transmit when the channel is perfectly secured or not. That is, we introduce the parameter α_k such that

$$f_k^{\text{mixed}} = \alpha_k f_k^{\text{sec}} + (1 - \alpha_k) f_k^{\text{conn}} \quad (31)$$

where $0 \leq \alpha_k \leq 1$ is the fraction of the time slots that Alice uses to send her data using the computational-only security schemes. If $\alpha_k = 1$, this means that Alice only transmits during the perfectly secured time slots. If $\alpha_k = 0$, this means that Alice transmits all the time, regardless of the channel security status (i.e., being in secrecy outage or not) and only

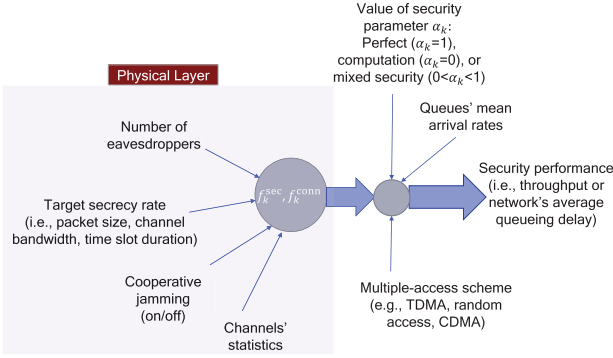


Fig. 4. Our proposed framework.

relies on the upper-layer security (cryptographic) schemes. The only change we need to make in our preceding analysis is to replace f_k^{conn} with f_k^{mixed} .

If the system was aided with strong cooperative jamming or the eavesdropper is far away from Alice such that its impact is negligible, then $f_k^{\text{sec}} = f_k^{\text{conn}}$ and the cryptographic security schemes might not be needed because the system will be always perfectly secured using the PHY security schemes. Since the exact CSI of the eavesdroppers is typically unknown, the exact perfectly secured time periods are not really known at Alice and, hence, both computational and perfect secrecy schemes are needed. Our proposed framework is depicted in Fig. 4.

IV. AVERAGE QUEUEING DELAY OF THE TDMA NETWORK UNDER SECURITY CONSTRAINTS

The longest queueing delay occurs when perfect secrecy is required. Additional long-delay cases can occur when the legitimate users (Alices) adopt approaches like on-off transmissions where Alice waits until her direct channel gain becomes high before transmission so that she can transmit. This is realized through a threshold on the channel gain of the Alice-Bob link. However, the transmission rate in such cases depends on the threshold imposed on the direct channel. As the security requirement (e.g., NSOP constraint) becomes more stringent, the user will have to wait for long times to start a data transmission session. Hence, in a multiple-access environment, it becomes less likely for a legitimate user to find a channel with high link gain and get assigned to it. Therefore, cooperative jamming schemes would be very helpful in such cases. Without cooperative jamming, the shortest queueing delay would occur when no secrecy constraints are imposed or when computational-only secrecy is implemented.

In such a multi-user system, our goal is to minimize the average queueing delay of the network which is given by

$$D_{\text{av}}^{\text{mixed}} = \frac{\sum_{k=1}^N \lambda_k D_k}{\sum_{k=1}^N \lambda_k} \quad (32)$$

where D_k is the queueing delay of the k -th Alice and is given by (48) in Appendix A. The optimization problem that minimizes the average queueing delay of the network is stated

as follows

$$\begin{aligned} \min : & \frac{\sum_{k=1}^N \lambda_k D_k}{\sum_{k=1}^N \lambda_k} \\ 0 \leq \{\omega_k\}_{k=1}^N \leq 1 & \\ = & \frac{\sum_{k=1}^N \lambda_k \frac{1-\lambda_k}{\mu_k - \lambda_k}}{\sum_{k=1}^N \lambda_k} = \sum_{k=1}^N \frac{e_k}{\omega_k f_k^{\text{mixed}} - \lambda_k} \\ \text{s.t. } & \mu_\ell > \lambda_\ell, \forall \ell, \\ & \sum_{\ell=1}^N \omega_\ell \leq 1 \end{aligned} \quad (33)$$

where $e_k = \frac{\lambda_k(1-\lambda_k)}{\sum_{k=1}^N \lambda_k}$ is a positive constant and $\mu_\ell > \lambda_\ell$ is the ℓ -th Alice's data queue stability constraint. Since the objective function is the sum of nonnegative convex functions and the constraints are linear in $(\omega_1, \omega_2, \dots, \omega_N)$, this optimization problem is a convex optimization program and can be solved using any convex optimization solver, e.g., the CVX toolbox [24]. Nevertheless, to gain further insights, we solve the two-user case analytically. In this case, using the mixed secrecy scheme, the optimization problem is given by

$$\begin{aligned} \min_{\omega_1, \omega_2} : & \frac{e_1}{\omega_1 f_1^{\text{mixed}} - \lambda_1} + \frac{e_2}{\omega_2 f_2^{\text{mixed}} - \lambda_2}, \\ \text{s.t. } & \frac{\lambda_1}{f_1^{\text{mixed}}} \leq \omega_1 \leq 1, \\ & \frac{\lambda_2}{f_2^{\text{mixed}}} \leq \omega_2 \leq 1, \\ & \omega_1 + \omega_2 = 1 \end{aligned} \quad (34)$$

The solution of this optimization problem is derived in Appendix B. The optimal solutions of ω_1 and ω_2 are given by

$$\omega_1^* = \frac{1 - \frac{\lambda_2}{f_2^{\text{mixed}}} + \sqrt{\frac{\lambda_2 \lambda_2}{\lambda_1 \lambda_1} \frac{\lambda_1}{f_1^{\text{mixed}}}}}{\sqrt{\frac{\lambda_2 \lambda_2}{\lambda_1 \lambda_1}} + 1} \quad (35)$$

and

$$\omega_2^* = 1 - \omega_1^* = 1 - \frac{1 - \frac{\lambda_2}{f_2^{\text{mixed}}} + \sqrt{\frac{\lambda_2 \lambda_2}{\lambda_1 \lambda_1} \frac{\lambda_1}{f_1^{\text{mixed}}}}}{\sqrt{\frac{\lambda_2 \lambda_2}{\lambda_1 \lambda_1}} + 1} \quad (36)$$

We note that the optimal value of the time-sharing vector is a nonlinear function of the arrival rates of the queues. If $\lambda_2 > \lambda_1$, as f_k^{mixed} ($k \in \{1, 2\}$) increases, the time assigned to user 1 decreases. This is because the system tries to maintain the stability of user 2 since its arrival rate is high, and to assign more time slots to the node with the higher arrival rate, which is user 2 in this case, hence, its average queueing delay decreases. The same conclusion holds for the converse case, i.e., the case of $\lambda_1 > \lambda_2$. In the case of symmetric users, where $\lambda_1 = \lambda_2$, the optimal time-sharing vector is $(0.5, 0.5)$.

Substituting with the optimal time-sharing vector (ω_1^*, ω_2^*) , the average queueing delay of the network is thus given by

$$D_{\text{av}}^{\text{mixed}} = \frac{\left(\sqrt{f_1^{\text{mixed}} \lambda_2 \lambda_2} + \sqrt{f_2^{\text{mixed}} \lambda_1 \lambda_1} \right)^2}{(\lambda_1 + \lambda_2)(f_1^{\text{mixed}} f_2^{\text{mixed}} - f_1^{\text{mixed}} \lambda_2 - f_2^{\text{mixed}} \lambda_1)} \quad (37)$$

If the channels are symmetric, $f_1^{\text{mixed}} = f_2^{\text{mixed}} = f^{\text{mixed}}$ and Eqn. (37) reduces to

$$D_{\text{av}}^{\text{mixed}} = \frac{\left(\sqrt{\lambda_2 \lambda_2} + \sqrt{\lambda_1 \lambda_1}\right)^2}{(\lambda_1 + \lambda_2)(f^{\text{mixed}} - \lambda_2 - \lambda_1)} \quad (38)$$

If perfect secrecy is required, f_k^{mixed} in all the expressions is replaced with f_k^{sec} , whose value is lower than f_k^{mixed} . Hence, the queuing delay becomes higher and a compromise between security and delay is required. Moreover, cooperative jamming can be employed to achieve both perfect security and low queuing delays.

The relationship between the average queuing delay under perfectly-secured transmissions, denoted by $D_{\text{av}}^{\text{sec}}$, and average queuing delay under computational-only secured transmissions, denoted by $D_{\text{av}}^{\text{conn}}$, is given by

$$\frac{D_{\text{av}}^{\text{conn}}}{D_{\text{av}}^{\text{sec}}} = \frac{f^{\text{sec}} - \lambda_2 - \lambda_1}{f^{\text{conn}} - \lambda_2 - \lambda_1} \quad (39)$$

Using the relationship between f^{sec} and f^{conn} , we get

$$\frac{D_{\text{av}}^{\text{conn}}}{D_{\text{av}}^{\text{sec}}} = \frac{\zeta f^{\text{conn}} - \lambda_2 - \lambda_1}{f^{\text{conn}} - \lambda_2 - \lambda_1} = \frac{\zeta - \frac{\lambda_2 + \lambda_1}{f^{\text{conn}}}}{1 - \frac{\lambda_2 + \lambda_1}{f^{\text{conn}}}} \quad (40)$$

with $\lambda_2 + \lambda_1 \leq \zeta f^{\text{conn}}$, which represents the queues stability condition. If $\zeta = 1$, the RHS will be equal to 1 and $D_{\text{av}}^{\text{conn}} = D_{\text{av}}^{\text{sec}}$.

If all queues have the same loads, i.e., $\lambda_k = \lambda$, we have

$$D_{\text{av}}^{\text{mixed}} = \frac{1 - \lambda}{\frac{f^{\text{mixed}}}{2} - \lambda} \quad (41)$$

which implies that $\omega_k = 1/2$ since $N = 2$ users. Also, the queuing delay in (41) is equal to the expression given by (48) in Appendix A which verifies our results.

For the general N -user case with symmetric loads and links' statistics, the network's average queuing delay is given by

$$\begin{aligned} D_{\text{av}}^{\text{mixed}} &= \frac{1 - \lambda}{\frac{f^{\text{mixed}}}{N} - \lambda} \\ &= \frac{1 - \lambda}{\frac{\alpha f^{\text{sec}} + (1 - \alpha) f^{\text{conn}}}{N} - \lambda} \\ &= \frac{1 - \lambda}{f^{\text{conn}} \frac{\alpha \zeta + (1 - \alpha)}{N} - \lambda} \end{aligned} \quad (42)$$

where $\alpha_k = \alpha$, $f_k^{\text{sec}} = f^{\text{sec}} = \zeta f^{\text{conn}}$, $f_k^{\text{conn}} = f^{\text{conn}}$, and $\omega_k = 1/N$. The minimum queuing delay occurs when $\alpha = 0$ and is given by

$$D_{\text{av}}^{\text{mixed}} = \frac{1 - \lambda}{\frac{f^{\text{conn}}}{N} - \lambda} \quad (43)$$

while the maximum queuing delay occurs at $\alpha = 1$ and is given by

$$D_{\text{av}}^{\text{mixed}} = \frac{1 - \lambda}{\frac{f^{\text{sec}}}{N} - \lambda} \quad (44)$$

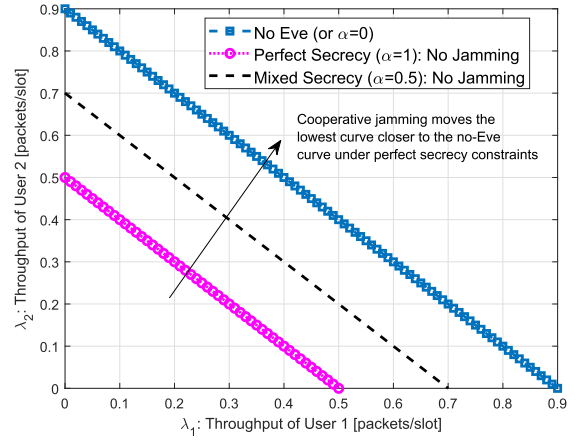


Fig. 5. Stable throughput region for no-Eve case (or computational security only case), perfect secrecy with no jamming (which we refer to as the secure stable-throughput region), and the mixed secrecy case where the transmitters still transmit data even if the channel is known to be imperfectly secured.

The condition on α that achieves bounded queuing delays at the users' queues is given by

$$\begin{aligned} \frac{\alpha f^{\text{sec}} + (1 - \alpha) f^{\text{conn}}}{N} &\geq \lambda \Rightarrow \alpha \leq \frac{f^{\text{conn}} - N\lambda}{f^{\text{conn}} - f^{\text{sec}}} \\ &= \frac{1 - N \frac{\lambda}{f^{\text{conn}}}}{1 - \zeta} \end{aligned} \quad (45)$$

with $\lambda \leq N f^{\text{conn}}$. If α is greater than $\frac{f^{\text{conn}} - N\lambda}{f^{\text{conn}} - f^{\text{sec}}} = \frac{1 - N \frac{\lambda}{f^{\text{conn}}}}{1 - \zeta}$, then the queuing delay is infinity.

V. NUMERICAL AND SIMULATION RESULTS

In this section, we present simulation results to corroborate our analysis. Unless otherwise stated explicitly, $N = 2$ users, $\lambda_k = 0.2$ packets/slot, $f_k^{\text{sec}} = 0.5$, $f_k^{\text{conn}} = 0.9$, and $\alpha_k = \alpha = 0.5$. First, we quantify the stable throughput without and with perfect secrecy and with mixed secrecy (where the system relies on both perfect secrecy and computational secrecy). In Fig. 5, we show the stable throughput when there is no Eve which represents the upper bound on performance. This case also arises when the legitimate system relies only on the computational secrecy schemes. Fig. 5 shows that the stable-throughput gap between the no-Eve case (i.e., $\alpha = 0$ or computational-only security case) and the perfect secrecy case which is given by $\zeta_k = \frac{f^{\text{sec}}}{f^{\text{conn}}}$. When cooperative jamming is used, as the jamming power increases, the curve of perfect secrecy moves closer to the no-Eve case and perfect secrecy is guaranteed. At those cases of strong cooperative jamming power, the system designer must set α to 1 and use PHY security to achieve perfect security. The mixed secrecy case is also shown in the figure to emphasize the fact that the throughput will increase when the perfect secrecy constraint is relaxed. Imposing the perfect secrecy constraint, reduces the maximum throughput from 0.9 packets/slot to 0.5 packets/slot with a total throughput loss up to 44%.

Matching our discussions in Section IV, and verified in Figs. 6 and 7, the longest queuing delay is incurred under perfect secrecy constraints, assuming knowledge of Eve's instantaneous CSI. To reduce the queuing delays, the system designer may relax the perfect secrecy constraints to a

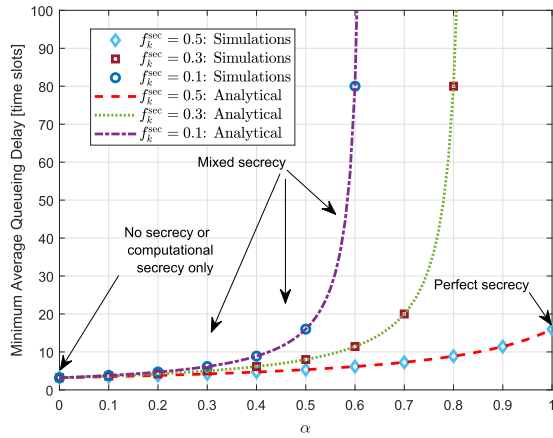


Fig. 6. Minimum average queuing delay of the network versus the mixed secrecy parameter, α .

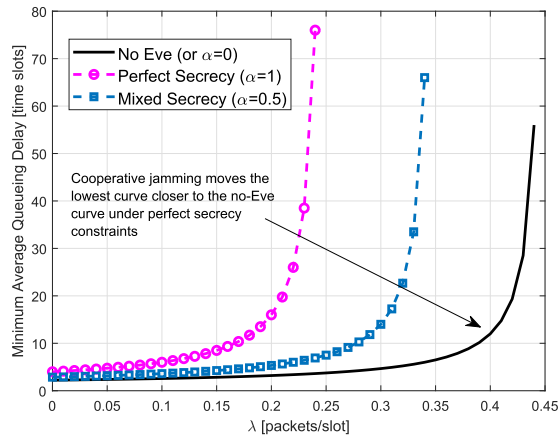


Fig. 7. Minimum average queuing delay of the network versus the input load $\lambda_1 = \lambda_2 = \lambda$ [packets/slot].

mixture between perfect secrecy and computational secrecy. The impact of the mixed secrecy parameter, denoted by α , is shown in Fig. 6 where we see that, when $f_k^{\text{sec}} = 0.5$, the queuing delay increases from 3.2 time slots to 16 time slots as α increases from 0 to 1. This shows the significant delay increase due to perfect secrecy constraints. If the system's average queuing delay is required to be no more than 3.2 time slots, the system designer has to set α to zero which implies that the transmissions are secured using computational-only security schemes. We emphasize that the 3.2 time slots average queuing delay can also be achieved under perfect secrecy using strong cooperative jamming power. For all the values of f_k^{sec} used to generate Fig. 6, all curves have the same queuing delay when $\alpha = 0$ since the computational-only security is utilized and the average queuing delay is affected only by f_k^{conn} , which is equal for all curves. When f_k^{sec} increases, which can happen when Eve moves closer to Alice, the queuing delay increases as α increases. Our proposed mixed security scheme can achieve any queuing delay between 3.2 and 16 time slots. In Fig. 7, we quantify the queuing delay versus the input load λ for the three cases of no Eve (or $\alpha = 0$), perfect secrecy (i.e., $\alpha = 1$), and mixed secrecy (i.e., $\alpha = 1/2$). When there are no perfect secrecy constraints, the queuing delay

is minimized since the legitimate user transmits whenever the channel is available. Under perfect secrecy constraints, the legitimate user needs to wait until the instantaneous secrecy rate allows the entire data packet to be transmitted perfectly secured. If the system designer relaxes the perfect secrecy constraints into allowing half (i.e., $\alpha = 1/2$) of the imperfectly secured time slots to be used for data transmission while using the computationally-secured cryptographic schemes for security, the queuing delay will be decreased significantly. For the cases of $f_k^{\text{sec}} = 0.3$ and $f_k^{\text{sec}} = 0.1$, if perfect security is used, the queuing delay will be infinity. Actually, the value of α that results in infinity queuing delay for each value of f_k^{sec} can be obtained from Eqn. (45).

With the help of cooperative jamming, if the legitimate user knows Eve's instantaneous CSI, the queuing delay curve will move from the perfect secrecy case closer to the no-Eve case. If Eve's instantaneous CSI is unknown at the Alices, an Alice should transmit whenever her link is not in outage and the jamming would be used for secrecy. In such cases, the queuing delay will always be equal to the no-Eve case. However, the attained secrecy is not perfect and it depends on the cooperative jamming parameters (i.e., jamming precoding matrix filter, cooperative jammer's transmit antenna, and the cooperative jammer's transmit power). Furthermore, if Alice decides to use on-off transmissions or wait to transmit until favorable conditions occur on her channel gain (i.e., the channel gain becomes higher than a certain threshold), the queuing delay would be worse than the perfect secrecy case.

VI. CONCLUSION

We derived insightful analytical relations between security and queuing delay and between security and throughput under eavesdropping attacks. First, we derived a simple relationship between the stable-throughput and the secure stable-throughput regions. Second, we quantified the impact of cooperative jamming on reducing the gap between the two regions in terms of a single key system parameter that depends on target secrecy rate and the distances between Alice, Bob, and Eve. Third, we derived the minimum queuing delay of the uplink system under security constraints and derived the optimal time-slot assignments. Fourth, we proposed a new scheme to combine PHY layer and computational secrecy and demonstrated its effectiveness in delay-limited scenarios. Under the perfect secrecy constraints, assuming symmetric links with identical statistics, we showed that the throughput loss due to eavesdropping is $1/(1 + 2^{\mathcal{R}})$, where $\mathcal{R}$ is the target secrecy rate, and the minimum throughput loss without cooperative jamming is 50%. Our numerical results demonstrated that, under perfect secrecy constraints, when $f_k^{\text{sec}} = 0.5$ and $f_k^{\text{conn}} = 0.9$, the maximum throughput decreases from 0.9 packets/slot to 0.5 packets/slot; i.e., a throughput loss of up to 44%. Moreover, the network's average queuing delay increases from 3.2 time slots to 16 time slots. This shows the significant queuing delay increase due to perfect secrecy constraints. We emphasize that the 3.2 time slots queuing delay is achieved with computational-only secrecy or with perfect secrecy under strong cooperative

jamming. Our proposed mixed security scheme can achieve any queuing delay value between 3.2 and 16 time slots and its achievable delay can be further decreased using cooperative jamming.

APPENDIX A QUEUE Q_k MARKOV CHAIN

Since each legitimate user, in a given time slot duration, can either transmit one information packet or receive one information packet (due to the Bernoulli arrival model), the Markov chain modeling the data queues at the Alices are modeled as *birth-death* processes. Consider the k -th Alice's data queue, Q_k . The probability that the queue transitions from state $n \in \{1, 2, \dots, \infty\}$ to state $n - 1$, denoted by $p_{k, \text{down}}^{n \rightarrow n-1}$, is given by the joint event that 1) the k -th Alice is selected for data transmission, which occurs with probability ω_k , 2) there is no arrived data packets at the queue, which occurs with probability $1 - \lambda_k$, and 3) the transmitted packet being decoded at Bob, which occurs with probability f_k^{conn} . Moreover, the probability that Q_k transitions from state $n \in \{1, 2, \dots, \infty\}$ to state $n + 1$, denoted by $p_{k, \text{up}}^{n \rightarrow n+1}$, is the probability of the joint event that 1) a data packet is arrived to the queue, which occurs with probability λ_k , and 2) the *complement* event that the legitimate user is scheduled for data transmission and the packet at the data queue head is decoded at Bob, which occurs with probability $1 - \mu_k = 1 - \omega_k f_k^{\text{conn}}$. If the data queue Q_k is in state 0, then the probability that the queue transitions to state $n = 1$, denoted by $p_{k, \text{up}}^{0 \rightarrow 1}$, is λ_k . Hence, the probability that the data queue state stays at the same state, increases by one packet, and the probability that the queue state decreases by one packet are given by

$$\begin{aligned} p_k^{0 \rightarrow 0} &= 1 - \lambda_k, \\ p_{k, \text{up}}^{0 \rightarrow 1} &= \lambda_k, \\ p_k^{n \rightarrow n} &= 1 - \lambda_k (1 - \mu_k) - (1 - \lambda_k) \mu_k \\ &= 1 - \lambda_k (1 - \omega_k f_k^{\text{conn}}) - (1 - \lambda_k) \omega_k f_k^{\text{conn}}, \\ p_{k, \text{up}}^{n \rightarrow n+1} &= \lambda_k (1 - \mu_k) = \lambda_k (1 - \omega_k f_k^{\text{conn}}), \\ p_{k, \text{down}}^{n \rightarrow n-1} &= (1 - \lambda_k) \mu_k = (1 - \lambda_k) \omega_k f_k^{\text{conn}} \end{aligned} \quad (46)$$

with $n \geq 1$.

Solving the state balance equations of the Markov chain modeling Q_k , the probability of the queue being in state $n \geq 0$, denoted by $\epsilon_{k, n}$, is given by

$$\begin{aligned} \epsilon_{k, n} &= \frac{\epsilon_{k, 0}}{1 - \mu_k} \left(\frac{(1 - \mu_k) \lambda_k}{(1 - \lambda_k) \mu_k} \right)^n, \\ \epsilon_{k, 0} &= \Pr\{Q_k = 0\} = 1 - \frac{\lambda_k}{\mu_k} \end{aligned} \quad (47)$$

where $\lambda_k < \mu_k$ represents the stability condition of queue Q_k . Using Little's law [25], the average queuing delay at Q_k is

$$D_k = \frac{\sum_{n=1}^{\infty} n \epsilon_{k, n}}{\lambda_k} = \frac{1 - \lambda_k}{\mu_k - \lambda_k} = \frac{1 - \lambda_k}{\omega_k f_k^{\text{conn}} - \lambda_k} \quad (48)$$

with $\mu_k > \lambda_k$ (stability condition).

APPENDIX B OPTIMAL ω_1 AND ω_2

The optimization problem in (34) can be solved as follows. Substituting by the equality constraint, $\omega_1 + \omega_2 = 1$, into the objective function, we get the following optimization problem in terms of ω_1

$$\begin{aligned} \min_{\omega_1} : & \frac{e_1/f_1^{\text{mixed}}}{\omega_1 - \frac{\lambda_1}{f_1^{\text{mixed}}}} + \frac{e_2/f_2^{\text{mixed}}}{1 - \omega_1 - \frac{\lambda_2}{f_2^{\text{mixed}}}}, \\ \text{s.t. } & \frac{\lambda_1}{f_1^{\text{mixed}}} \leq \omega_1 \leq 1, \frac{\lambda_2}{f_2^{\text{mixed}}} \leq 1 - \omega_1 \leq 1 \end{aligned} \quad (49)$$

Taking the first derivative of the new objective function with respect to ω_1 , we get

$$-\frac{e_1/f_1^{\text{mixed}}}{(\omega_1 - \frac{\lambda_1}{f_1^{\text{mixed}}})^2} + \frac{e_2/f_2^{\text{mixed}}}{(1 - \frac{\lambda_2}{f_2^{\text{mixed}}} - \omega_1)^2} \quad (50)$$

Equating the first derivative to zero, we get

$$\frac{e_1/f_1^{\text{mixed}}}{(\omega_1^* - \frac{\lambda_1}{f_1^{\text{mixed}}})^2} = \frac{e_2/f_2^{\text{mixed}}}{(1 - \frac{\lambda_2}{f_2^{\text{mixed}}} - \omega_1^*)^2} \quad (51)$$

where ω_1^* is the equation root. Taking the square root of both sides of the equation, and excluding the invalid root, we get

$$\sqrt{e_1/f_1^{\text{mixed}}}(1 - \frac{\lambda_2}{f_2^{\text{mixed}}} - \omega_1^*) = \sqrt{e_2/f_2^{\text{mixed}}}(\omega_1^* - \frac{\lambda_1}{f_1^{\text{mixed}}}) \quad (52)$$

After some mathematical manipulations, we get

$$(1 - \frac{\lambda_2}{f_2^{\text{mixed}}} - \omega_1^*) = \frac{\sqrt{e_2/f_2^{\text{mixed}}}}{\sqrt{e_1/f_1^{\text{mixed}}}}(\omega_1^* - \frac{\lambda_1}{f_1^{\text{mixed}}}) \quad (53)$$

Thus,

$$\omega_1^* = \frac{1 - \frac{\lambda_2}{f_2^{\text{mixed}}} + \frac{\sqrt{e_2/f_2^{\text{mixed}}}}{\sqrt{e_1/f_1^{\text{mixed}}}} \frac{\lambda_1}{f_1^{\text{mixed}}}}{\frac{\sqrt{e_2/f_2^{\text{mixed}}}}{\sqrt{e_1/f_1^{\text{mixed}}}} + 1} \quad (54)$$

and

$$\omega_2^* = 1 - \omega_1^* = 1 - \frac{1 - \frac{\lambda_2}{f_2^{\text{mixed}}} + \frac{\sqrt{e_2/f_2^{\text{mixed}}}}{\sqrt{e_1/f_1^{\text{mixed}}}} \frac{\lambda_1}{f_1^{\text{mixed}}}}{\frac{\sqrt{e_2/f_2^{\text{mixed}}}}{\sqrt{e_1/f_1^{\text{mixed}}}} + 1} \quad (55)$$

Substituting with the values of e_1 and e_2 , we get the expressions in (35) and (36).

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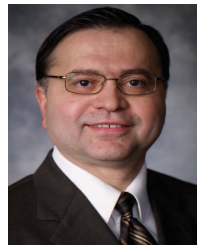
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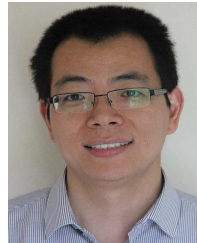


Ahmed El Shafie (SM'18) received the B.Sc. degree (Hons.) in electrical engineering from Alexandria University, Alexandria, Egypt, in 2009, the M.Sc. degree in communication and information technology from Nile University, Cairo, Egypt, in 2014, and the Ph.D. degree from The University of Texas at Dallas, Richardson, TX, USA, in 2018. Since 2018, he has been a Senior Systems Engineer with Qualcomm Inc., San Diego, CA, USA. He was a recipient of the David Daniel Best Doctoral Thesis Award in 2018, and the Jonsson School Industrial

Advisory Council Fellowship Award in 2017. He received the IEEE TRANSACTIONS ON COMMUNICATIONS Exemplary Reviewer Award in 2015, 2016, and 2017. He also received the IEEE COMMUNICATIONS LETTERS Exemplary Reviewer Award in 2016. He is nominated for the 2018 CGS/ProQuest Distinguished Dissertation Award. He currently serves as an Editor for the IEEE COMMUNICATIONS LETTERS, *Physical Communications*, and the *Transactions on Emerging Technologies in Telecommunications*. In addition, he serves as a Guest Editor for the IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING.



Naofal Al-Dhahir received the Ph.D. degree in electrical engineering from Stanford University. From 1994 to 2003, he was a Principal Member of the Technical Staff at GE Research and the AT&T Shannon Laboratory. He is currently the Erik Jonsson Distinguished Professor with The University of Texas at Dallas. He is the co-inventor of 41 issued U.S. patents, a co-author of over 400 papers, and a co-recipient of four IEEE Best Paper Awards. He is an Editor-in-Chief of the IEEE TRANSACTIONS ON COMMUNICATIONS.



Zhiguo Ding received the B.Eng. degree in electrical engineering from the Beijing University of Posts and Telecommunications in 2000 and the Ph.D. degree in electrical engineering from Imperial College London in 2005. From 2005 to 2018, he was with Queen's University Belfast, Imperial College London, Newcastle University, and Lancaster University. From 2012 to 2018, he was an Academic Visitor with Princeton University. Since 2018, he has been with The University of Manchester as a Professor in communications. His research interests

include 5G networks, game theory, cooperative and energy harvesting networks, and statistical signal processing. He received the Best Paper Award at IET ICWMC-2009 and IEEE WCSP-2014, the EU Marie Curie Fellowship 2012–2014, the Top IEEE TVT Editor 2017, the IEEE Heinrich Hertz Award 2018, and the IEEE Jack Neubauer Memorial Award 2018. He is serving as an Editor for the IEEE TRANSACTIONS ON COMMUNICATIONS, the IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, and the *Journal of Wireless Communications and Mobile Computing*. He was an Editor of the IEEE WIRELESS COMMUNICATION LETTERS and the IEEE COMMUNICATION LETTERS from 2013 to 2016.



Ridha Hamila received the M.Sc. degree, the Licentiate of Technology degree (Hons.), and the Doctor of Technology degree from the Department of Information Technology, Tampere University of Technology (TUT), Tampere, Finland, in 1996, 1999, and 2002, respectively. From 1994 to 2002, he held various research and teaching positions with the Department of Information Technology, TUT. From 2002 to 2003, he was a System Specialist with the Nokia Research Center and Nokia Networks, Helsinki. From 2004 to 2009, he was with the Etisalat University College, Emirates Telecommunications Corporation, United Arab Emirates. He served as a Supervisor for a large number of under/graduate students and post-doctoral fellows. He is currently an Associate Professor with the Department of Electrical Engineering, Qatar University, Qatar. He is also an Adjunct Professor with the Department of Communications Engineering, TUT. His current research interests include mobile and broadband wireless communication systems, cellular and satellites-based positioning technologies, and synchronization and DSP algorithms for flexible radio transceivers. He has been involved in several past and current industrial projects Qtel, QNRF, Finnish Academy projects, TEKES, Nokia, and EU research and education programs.